MIL-M-46317A(Ord)
28 June 1962
SUPERSEDING
MIL-M-46317(Ord)
16 June 1961

MILITARY SPECIFICATION

MOUNT, TELESCOPE: XM117

1. SCOPE

1.1 This specification covers one type of telescope mount which supports Elbow Telescope XM107 and provides means for measuring azimuth, elevation and angle of sight readings in mils (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment, and Tools; Packaging of
MIL-I-45208	Inspection Requirements, General Specification For
MIL-I-45607	Inspection Equipment, Supply and Maintenance for Ordnance

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

F8579494	Mount, Telescope: XM117

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DRAWINGS (Cont'd)

LSQAP 8579494	Supplementary Quality Assurance Provision List
IEL 8579494	Index of Inspection Equipment List

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one mount, representative of production, shall be made available to the Government to determine if the mount meets all the requirements of this specification.

3.2 Materials.- The materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction.- The mount shall be manufactured in accordance with Drawing F8579494, "Mount, Telescope: XM117" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to assure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 Operating temperature.- The mount shall be operable and shall meet the requirements specified in 3.6.7, 3.6.8, and 3.6.9 while exposed to and thermally stabilized at ambient temperatures of plus 150 and minus 65 degrees Fahrenheit.

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3.5.2 Vibration.- The mount shall meet all requirements of this specification after being vibrated in a vertical plane in the normal operating position for five minutes, plus or minus 15 seconds at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second. During vibration the mount shall support an Elbow Telescope XM107 or equivalent in size and weight.

3.6 Orientation.- All performance requirements shall be met with the dovetail of the mount positioned as follows:

(a) The dovetail shall be in a vertical plane in accordance with the test procedure as outlined in 4.9.3.4.

(b) The zero setting of the angle of site, elevation and azimuth scales, including respective micrometer scales shall be in coincidence with their respective indices.

3.6.1 Level vials.

3.6.1.1 Level vial settings.- The elevation and cross-level vial bubbles shall be centered within the thickness of one-half a graduation space on the level vial.

3.6.1.2 Elevation.- The elevation bubble reading shall not vary by more than one-half of a graduation space on the level vial as a result of canting the mount 10 degrees to the right and left.

3.6.1.3 Cross level.- The cross level bubble reading shall not vary by more than one-half of a graduation space on the level vial while rotating the mount plus and minus 200 mils about the elevation axis.

3.6.2 Plumb travel.- Utilizing an Elbow Telescope XM107 or equivalent in the mount, and the mount positioned in accordance with 3.6 with the optical line of sight in a horizontal plane, the optical line of sight shall not deviate from a plumb line by more than plus or minus 0.5 mil (total lateral deviation from plumb line not to exceed 0.5 mil) at any azimuth setting when the line of sight of the telescope, or equivalent, is elevated and depressed from the horizontal as follows:

MOVEMENT	ELEVATION	DEPRESSION
Rotation about angle of sight axis	200 mils	200 mils
Rotation about elevation axis of mount	800 mils	300 mils

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3.6.2.1 Side play.- In reversing the elevation motion by means of the elevation micrometer knob and angle of sight micrometer knob, the vertical displacement of the line of sight from the plumb line shall not exceed 0.2 mil.

3.6.3 Horizontal travel.- Utilizing an Elbow Telescope XM107 or equivalent in the mount and the mount positioned in accordance with 3.6, the optical line of sight shall not vary in elevation by more than 1 mil (total excursion not to exceed 1 mil) while moving the mount 360 degrees in azimuth.

3.6.3.1 Lift.- In reversing the azimuth motion by means of the micrometer knob the displacement between the line of sight and a horizontal line shall not exceed 0.2 mil.

3.6.4 Open sight.- The line of the sight of the open sight (Posts 8579591 and 8579595) shall be parallel to the optical line of sight of the Elbow Telescope XM107 or equivalent, within 10 mils in azimuth and elevation.

3.6.5 Angular accuracy.- With the mount positioned in accordance with 3.6, the following requirements shall be met.

3.6.5.1 Circle error.- Azimuth accuracy readings from zero position shall be taken in 800 mil increments throughout 360 degrees travel by means of rotating the micrometer knob, and the scale readings shall indicate the true angular deviation within an accuracy of 0.5 mil excluding backlash at each reading.

3.6.5.2 Elevation error.- Elevation accuracy readings from zero position shall be taken in 200 mil increments throughout maximum elevation and depression by means of rotating the elevation micrometer knob, and the elevation scale readings shall indicate the true angular deviation within an accuracy of 0.7 mil excluding backlash at each reading.

3.6.5.3 Angle of site error.- The angle of site accuracy readings from zero position shall be taken in 100 mil increments throughout maximum elevation and depression by means of rotating the angle of site micrometer knob, and the angle of site scale readings shall indicate the true angular deviation within an accuracy of 0.7 mil excluding backlash at each reading.

3.6.6 Backlash.- Backlash in the elevation and azimuth worm and worm wheel mechanisms shall not exceed 0.5 mil.

3.6.6.1 Backlash in the angle of site mechanism shall not exceed 0.5 mil.

4.

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3.6.7 Dovetail locking device.- The dovetail locking device shall be capable of firmly securing the mount to a dovetail plate (which mounts on Telescope Mount Holder XM5 and XM6, or equivalent in size and weight). When the locking device is actuated and released, the mount shall be removable without force or restriction when subjected to operating temperatures specified in 3.5.1.

3.6.8 Torque.- The torque required to operate the knobs and scales at standard ambient temperature (60 to 90 degrees Fahrenheit) shall be within the tolerances specified in Table I. When measured at the extreme temperatures specified in 3.5.1, the torque values shall not have changed from the respective values obtained at room temperature by more than 50 percent.

TABLE I

TORQUE VALUES IN OUNCE INCHES

	MINIMUM	MAXIMUM
Elevation Knob	32	80
Azimuth Knob	32	80
Angle of Site Knob	4	32

3.6.9 Telescope clamping mechanism.- The mount shall firmly secure an Elbow Telescope XM107 or equivalent in size and weight, when actuating the telescope clamping mechanism at the operating temperatures specified in 3.5.1.

3.7 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

3.8 Support dimension.- The support dimension of 0.843 specified on Drawing F8579499 shall be capable of accepting the elbow telescope.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements.- Inspection shall be in accordance with Specification MIL-I-45208.

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4.3 Production model.- Unless otherwise specified in the contract, one mount shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.1 Defective units.- Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.5 Inspection of components.- Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP)-8579494.

4.5.1 Contents of SQAP.- The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual non-conforming items.

4.5.2 Contractor's inspection for all components not covered by SQAP.- Inspection for these components shall be in accordance with the inspection procedure of the written inspection system required by Specification MIL-I-45208. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the telescope mount.

4.6 Material.- Control for requirements of material shall be in accordance with Test Data requirements of Specification MIL-I-45208.

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4.7 Inspection equipment.- Inspection equipment shall conform to drawings listed in IEL-8579494, except as provided for under the alternative inspection procedures and inspection equipment requirement of Specification MIL-I-45208. Provisions of Specification MIL-I-45607 shall apply to test equipment provided by the Government or provided by the contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired for verification inspection.

4.7.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except in cases where Ordnance drawings are applicable. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8 Tests.

4.8.1 Classification of defects.- The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.8.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L7 of Table IIIB with Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: None defined

MAJOR: AQL 4.0%	Req't Par.	QA. Par.	Test Method
101. Operating temperatures	3.5.1	4.9.1	Cold Chest & Oven
102. Vibration	3.5.2	4.9.2	Standard Measuring Equipment (SME)

MINOR: None defined

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TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table IIIA with Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: None defined

MAJOR: AQL 2.5%		Req't Par.	QA. Par.	Test Method
101.	Orientation	3.6	4.9.3	Special Testing Equipment(STE) Gage
102.	Level vial settings	3.6.1.1	4.9.3.5	STE-Gage
103.	Elevation	3.6.1.2	4.9.3.6	STE-Gage
104.	Cross Level	3.6.1.3	4.9.3.7	STE-Gage
105.	Plumb travel	3.6.2	4.9.3.8	STE-Gage
106.	Side play	3.6.2.1	4.9.3.9	Visual
107.	Horizontal travel	3.6.3	4.9.3.10	STE-Gage
108.	Lift	3.6.3.1	4.9.3.11	Visual
109.	Open sight	3.6.4	4.9.3.12	Visual-Chart
110.	Circle error	3.6.5.1	4.9.3.13	STE-Gage
111.	Elevation error	3.6.5.2	4.9.3.14	STE-Gage
112.	Angle of site error	3.6.5.3	4.9.3.15	STE-Gage
113.	Backlash	3.6.6	4.9.3.16	Visual
114.	Dovetail locking	3.6.7	4.9.3.17	Manual-STE
115.	Torque	3.6.8	4.9.3.18	STE
116.	Telescope clamping mechanism	3.6.9	4.9.3.19	Manual-STE
117.	Identification marking	3.7	4.9.3.20	Visual
118.	Support dimension .843	3.8	4.9.3.21	Standard Meas. Equipment(SME)

Minor: None defined

4.9 Test methods and procedures.

4.9.1 Operating temperature test.- Operating temperature test consists of exposing the mount to temperatures indicated in 3.5.1 by means of a cold chest and an oven capable of maintaining these temperatures until the mount is thermally stabilized. While at these temperatures, the mount shall be checked for all requirements of 3.5.1.

4.9.2 Vibration.- The mount shall be vibrated as indicated in 3.5.2 by means of a vibrating machine on which the mount is supported by a device for holding the mount in the normal operating position. The vibrating machine shall be capable of vibrating the mount at the number of cycles, amplitude and time for the requirements in 3.5.2. Upon completion of this test, the mount shall be visually inspected for evidence of damage and then shall meet all performance requirements of this specification.

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4.9.3 Performance tests.- Performance tests are accomplished by means of an elevation and cross level fixture, final inspection and collimation fixture, perpendicularity gage, azimuth test fixture and adapter assembly. These fixtures and gages are depicted on Drawings D23553, D6510966, D7646979, D7691596 and C7651341, respectively, listed on IEL-8579494.

4.9.3.1 Elevation and cross level fixture.- Elevation and cross level fixture provides the means for elevating and depressing the mount in a vertical plane, and also provides the mounting plate for attaching and locating the final inspection and collimating fixture.

4.9.3.2 Final inspection and collimation fixture.- Final inspection and collimation fixture is an adapter fixture which provides two dovetail slots 90 degrees apart for positioning the mount. It is attached to the mounting plate of the elevation fixture and located so that the axis of the elevation mechanism of the mount coincides with the axis of the elevation fixture.

4.9.3.3 Perpendicularity gage.- Perpendicularity gage is designed to fit into the dovetail slot of the final inspection fixture and is provided with a flat top surface perpendicular to the flat surface of the dovetail slot. A "V" slot is machined to the top surface to accommodate a collimating telescope.

4.9.3.4 Set-up for performance tests.- The set-up for performance tests is made by cross leveling the elevation fixture by means of a ten second bench level with the elevation scale set at zero. The final inspection fixture is then attached to the mounting plate of the elevation fixture and cross leveled by means of pads provided for this purpose. Both dovetail slots are then checked by means of the perpendicularity gage. In this position, the dovetail slots will be in a vertical plane as required in 3.6(a). The telescope mount is now inserted in place and the zero settings are visually set as indicated in paragraph 3.6(b).

4.9.3.5 Level vial settings.- Level vial settings are visually checked for the tolerances of 3.6.1.1 when the mount is positioned as required in 4.9.3.4.

4.9.3.6 Elevation test.- Elevation test is made by using the dovetail slot, the flat surface of which is parallel to the axis of the elevation fixture. With the mount positioned as described in 4.9.3.4, an excursion is made from minus 10 degrees to plus 10 degrees by movement of the elevation fixture. During the entire excursion, the elevation bubble is observed and shall not exceed the requirements of 3.6.1.2.

4.9.3.7 Cross level check.- Cross level check consists of positioning the mount in the vertical position and observing the cross level bubble while moving the elevation knob plus and minus 200 mils. The bubble movement shall not exceed the error of 3.6.1.3.

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4.9.3.8 Plumb travel test.- Plumb travel test is made by using the dovetail slot which has the flat surface perpendicular to the axis of the elevation fixture. A plumb line is located approximately fifty feet from the fixture and in alinement with a collimating telescope placed in the "V" slot of the perpendicularity gage located in the dovetail slot. This will establish the line of sight for the Elbow Telescope XM107 or equivalent of the mount. The mount is then placed in the dovetail slot and positioned as indicated in 4.9.3.4. A peep sight with a 1/8 inch hole placed over the eyepiece of the Elbow Telescope XM107 or equivalent will assist the observer's eye. The angle of sight and elevation knobs are then checked for lateral deviation and tolerance as indicated in 3.6.2. In cases where the plumb line is not long enough for a full excursion, the elevation fixture will compensate for full travel. A mil block chart located at the top or bottom of the plumb line will indicate the error for this test.

4.9.3.9 Side play check.- Side play check is made as indicated in 3.6.2.1 in conjunction with a mil block chart located in back of the plumb line.

4.9.3.10 Horizontal travel test.- Horizontal travel test is performed with the aid of the azimuth test fixture and adapter assembly. The adapter assembly is attached to the mounting plate of the azimuth test fixture and provides the dovetail slot for positioning the mount. With the azimuth fixture cross leveled, the dovetail slot is positioned in a vertical plane with the perpendicularity gage. The test consists of checking the line of sight of the Elbow Telescope XM107 or equivalent at 0, 90, 180 and 270 degrees of the azimuth test fixture for deflection in a horizontal plane of 360 degrees excursion of the mount. The error shall not exceed the requirements of 3.6.3.

4.9.3.11 Lift check.- Lift check is made at the four positions of the azimuth test fixture as described above in the horizontal travel test and shall not exceed the requirement of 3.6.3.1.

4.9.3.12 Open sight test.- Open sight test is made by means of a wall chart located approximately fifty feet from the objective of the Elbow Telescope XM107 or equivalent. The chart shall have a mark for the line of sight of the telescope, and a ten mil radius circle the center of which will correspond to the center of the open sight. With the telescope focused on the line of sight mark, the open sight shall read within the limits of the circle.

4.9.3.13 Circle error test.- Circle error test is made with the same set-up employed in the horizontal travel test of 4.9.3.10. With the mount and azimuth fixture set at zero and in alinement with the target collimator of the fixture, readings are taken in 800 mil increments through 360 degrees of travel and the deviation shall not exceed the requirements of 3.6.5.1.

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4.9.3.14 Elevation error.- Elevation error is checked using the same set-up as used for the plumb travel test of 4.9.3.8. In addition, a horizontal zero mark shall be located on a wall chart in back of the plumb line and in a horizontal plane with the optical line of sight of the Elbow Telescope XM107, or equivalent. A mil block located above the line and one below will indicate the error for this test. With the mount located in the dovetail of the fixture, the zero line of the telescope shall coincide with the zero mark of the wall chart. The elevation fixture is elevated to 11 degrees 15 minutes (200 mils) and then the elevation knob of the mount moved 200 mils which should coincide the zero line of the telescope with the horizontal mark of the wall target. Deviation can be read on the mil blocks and shall be within the tolerance of 3.6.5.2. This operation shall be repeated for the entire scale reading.

4.9.3.15 Angle of site error test.- Angle of site error test is made using the same equipment and method as used for the elevation error in 4.9.3.14. The 100 mil increment requirements of 3.6.5.3 are equivalent to 5 degrees 37-1/2 minutes as measured on the elevation fixture scale.

4.9.3.16 Backlash.- Backlash shall be checked at positions where accuracy readings are taken on the targets described for these tests. It consists of sighting the Elbow Telescope XM107, or equivalent, installed on the mount on the target with a clockwise movement of each micrometer knob and then with a counterclockwise movement, and the difference in readings of each movement is considered as backlash. The tolerance for these tests shall not exceed the requirements of 3.6.6 and 3.6.6.1.

4.9.3.17 Dovetail locking device.- Dovetail locking device is tested for operability when subjected to temperatures specified in 3.5.1. Final inspection and collimation fixture is used for testing the dovetail locking device.

4.9.3.18 Torque test.- Torque test consists of measuring the torque required to rotate the three knobs of the mount through full travel at room temperature by means of a commercial torque wrench with a special adapter. These torque values are then compared with operating temperatures test of 3.5.1 and the difference shall not exceed the error of 3.6.8.

4.9.3.19 Telescope clamping mechanism.- Telescope clamping mechanism is tested manually at temperatures specified in 3.5.1 for operability. When clamped, the Elbow Telescope XM107, or equivalent, should be firmly held in all positions and when released shall have a free movement at all positions.

4.9.3.20 Identification marking test.- Identification marking test is visual and made in accordance with Standard MIL-STD-130.

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4.9.3.21 Support dimension.- The dimension .843 inch specified on Drawing F8579499 shall be checked using standard measuring equipment for assurance that it will receive the telescope.

4.9.3.22 Preservation, packaging, packing and marking.- The quality assurance provisions for the preservation, packaging, packing and marking for shipment shall be in accordance with the applicable document specified in 5.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The mount shall be packaged, packed and marked in accordance with Specification MIL-F-14619/P12073. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- Telescope Mount XM117 and Elbow Telescope XM107 are used for direct and indirect firing of Battle Group Weapons System XM28 and XM29.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Selection of applicable levels of preservation, packaging, and packing (see 5.1)
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Design information.- Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Special testing equipment (STE).- Special testing equipment is defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.4.2 Standard measuring equipment (SME).- Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, and so forth.

6.4.3 Production model.- A model in its final mechanical form of final production design made by production methods, tools, jigs, and fixtures.

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